

Electronic Devices & Circuits Laboratory
Experiment 4: MOSFET Differential Amplifier — DC Bias, Common-Mode and Differential-Mode Transfer Characteristics

1. Objective

To assemble a simple two-transistor MOSFET differential amplifier using one SI9926BDY dual N-channel MOSFET package, to set and verify its DC operating point, to experimentally obtain the common-mode transfer characteristic and input common-mode range (ICMR), and the differential-mode transfer characteristic (ΔI_D versus V_{id}); and to compare the measured small-signal transconductance with the theoretical prediction.

2. Components and Equipment Required

Item	Specification	Quantity
MOSFET	SI9926BDY, dual N-channel, SO-8, with SO-8-to-DIP breakout adapter	1
Drain resistors RD1, RD2	130 Ω , $\pm 5\%$, ≥ 1 W (2 W preferred — dissipation approaches 0.8 W at full differential swing)	2
Tail resistor RSS	12 Ω , $\pm 5\%$, $\frac{1}{4}$ W or higher	1
Potentiometer	10 k Ω , multi-turn, for V_{in+} and V_{in-} bias adjustment	2
DC power supply	0–30 V regulated, current-limited	1
Digital multimeter	For V_{D1} , V_{D2} , V_{RSS} measurement	2–3
Breadboard and jumper wires	—	as required
Datasheet	SI9926BDY, Vishay Siliconix (provided copy)	1

3. Pin Configuration (SO-8, top view)

Confirm the following pin assignment against the physical marking on the breakout board and the provided datasheet before wiring; adapter silkscreens can occasionally differ from the die pinout.

Pin	Terminal	Belongs to
1	S1 (Source)	M1
2	G1 (Gate)	M1
3	S2 (Source)	M2
4	G2 (Gate)	M2
5, 6	D2 (Drain)	M2
7, 8	D1 (Drain)	M1

4. Theory

Two matched N-channel MOSFETs (M1, M2 — the two dice inside one SI9926BDY package) have their sources tied to a common node biased by a tail resistor RSS, and their drains loaded by equal resistors RD1 = RD2 = RD to VDD. The gates form the differential inputs Vin+ and Vin-. Because the two transistors share a tail node, their drain currents satisfy ID1 + ID2 = ISS at all times, where ISS is the current flowing into RSS.

For an ideal square-law MOSFET in strong inversion, the differential-mode large-signal transfer characteristic is:

$$\Delta ID = ID1 - ID2 = \beta \cdot Vid \cdot \sqrt{(ISS/\beta - Vid^2/4)}, \text{ valid for } |Vid| < Vid_{max} = \sqrt{(2 \cdot ISS/\beta)}$$

where Vid = Vin+ - Vin- and $\beta = k'n(W/L)$ is the transistor's transconductance parameter. This model needs a gate overdrive VOV of roughly 100 mV or more to hold well.

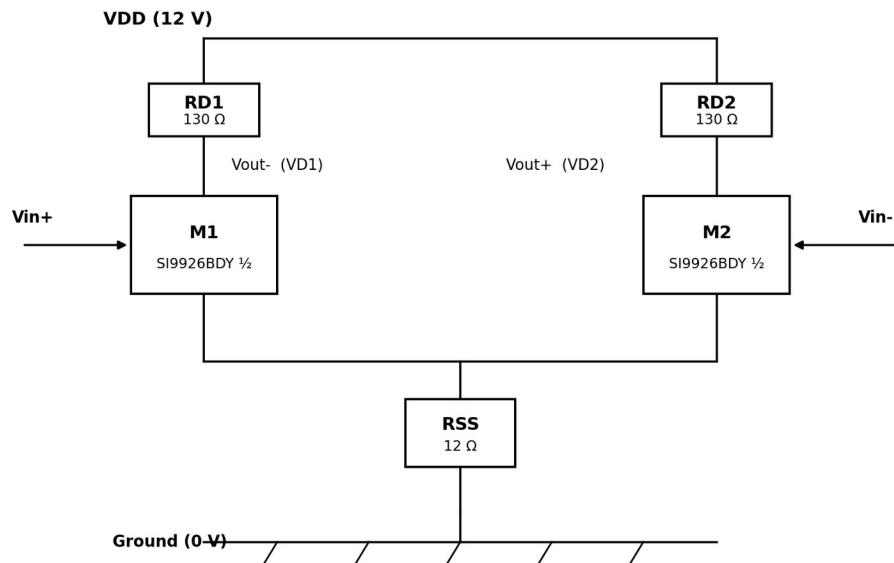
Bias current for this device: SI9926BDY is a power MOSFET with a large β ($\approx 51 \text{ A/V}^2$, computed from its own datasheet $gfs = 29 \text{ S}$ rating). Reaching VOV $\geq 100 \text{ mV}$ with this device would require a bias current well into the amp range — impractical for a bench differential-pair demonstration kept under 100 mA. At the ISS $\approx 80 \text{ mA}$ ($\approx 40 \text{ mA}$ per branch) used in this experiment, the square-law estimate gives VOV0 $\approx 40 \text{ mV}$, which is well below that threshold: the pair is operating in weak inversion, not strong inversion. This is a genuine, honest property of this device at low current, not a simplification — the square-law formula above is given for reference but is not expected to fit the measured curve here.

In weak inversion, MOSFET drain current follows an exponential law analogous to a BJT, $ID \propto \exp(VGS/nVT)$, where $VT = kT/q \approx 25.85 \text{ mV}$ at room temperature and n is a process-dependent slope factor (typically 1.2-2 for this class of device). For the differential pair this gives:

$$\Delta ID = ISS \cdot \tanh(Vid / 2nVT), \quad gm = ISS / (nVT), \quad Ad \approx gm \cdot RD$$

The common-mode transfer characteristic (VD1, VD2 versus Vcm at Vid = 0) stays flat only while both transistors remain in saturation and RSS behaves as a good current source; its flat region defines the input common-mode range (ICMR). A finite RSS also sets a finite common-mode gain, $A_{cm} \approx -RD/(2 \cdot RSS)$, giving CMRR = $|Ad/A_{cm}| \approx gm \cdot RSS$.

5. Circuit Diagram



6. Safety Precautions

- The SI9926BDY has a low gate threshold ($V_{GS(th)} = 0.6\text{--}1.5\text{ V}$, min-max per datasheet) and, like all TrenchFET devices, is sensitive to electrostatic discharge. Ground yourself before handling the breakout board.
- Set the power supply current limit to about 150 mA before connecting the circuit for the first time — comfortably above the 80 mA target but low enough to catch wiring faults.
- Do not exceed $V_{GS} = \pm 12\text{ V}$ or $V_{DS} = 20\text{ V}$ under any circumstance.
- Determine I_S and the branch currents from voltage measurements across R_{SS} and R_D (as in the procedure) rather than inserting a multimeter in series, to keep the measurement non-invasive and consistent across readings.
- Verify all pin connections twice against the datasheet before switching on the supply.
- Always power off before changing any wiring on the breadboard.

7. Procedure

7.1 DC bias-point setting

- Assemble the circuit as shown in Section 5, with $V_{DD} = 12\text{ V}$, $R_{D1} = R_{D2} = 130\ \Omega$, $R_{SS} = 12\ \Omega$.
- Set both input potentiometers to mid-scale. Switch on the supply with the current limit set to about 150 mA.
- Adjust both pots together, keeping $V_{in+} = V_{in-} = V_{cm}$, while monitoring the voltage across R_{SS} , until $I_S = V_{RSS} / R_{SS} \approx 80\text{ mA}$ ($V_{RSS} \approx 0.96\text{ V}$). Given the datasheet's $0.6\text{--}1.5\text{ V}$ threshold spread, V_{cm} will typically land somewhere in the

range 1.6–2.5 V — the exact value depends on the actual device and is not critical, only the resulting I_S is.

- Record V_{cm} , V_{D1} and V_{D2} at this quiescent (Q) point. The difference $V_{D1} - V_{D2}$ at $V_{id} = 0$ indicates the inherent device mismatch of the pair.

7.2 Common-mode transfer characteristic

- Keeping $V_{in+} = V_{in-}$ at all times ($V_{id} = 0$), vary V_{cm} in steps of 0.5 V from about 1.0 V upward.
- At each step, record V_{cm} , V_{D1} , V_{D2} and V_{RSS} (hence I_S).
- Stop when V_{D1} or V_{D2} comes within about 0.5 V of V_{DD} (transistor entering triode) or when I_S collapses toward zero (transistor entering cutoff). These two limits mark the ICMR.

7.3 Differential-mode transfer characteristic

- Set V_{cm} to the Q-point value found in Section 7.1. Keep V_{in-} fixed at this value throughout.
- Vary V_{in+} in steps of about 25 mV over a range of ± 300 mV around V_{cm} ; record the exact V_{in+} at each step.
- At each step, record V_{D1} and V_{D2} , and compute $I_{D1} = (V_{DD} - V_{D1})/R_D$, $I_{D2} = (V_{DD} - V_{D2})/R_D$, and $\Delta I_D = I_{D1} - I_{D2}$, against $V_{id} = V_{in+} - V_{cm}$.
- Repeat symmetrically for V_{in+} below V_{cm} (negative V_{id}).

8. Observation Tables

Table 1 — Common-mode sweep ($V_{id} = 0$)

V_{cm} (V)	V_{D1} (V)	V_{D2} (V)	V_{RSS} (V)	$I_S = V_{RSS}/R_{SS}$ (mA)
...	...	...	...	...

Table 2 — Differential-mode sweep (V_{cm} held at Q-point)

V_{in+} (V)	$V_{id} = V_{in+} - V_{cm}$ (mV)	V_{D1} (V)	V_{D2} (V)	I_{D1} (mA)	I_{D2} (mA)	ΔI_D (mA)
...	...	...	...	...	...	...

9. Graphs to Be Plotted

- V_{D1} and V_{D2} versus V_{cm} — common-mode transfer characteristic; mark the ICMR limits.
- ΔI_D versus V_{id} — on linear axes.
- $|\Delta I_D|$ versus V_{id} — on semi-log axes (log Y, linear X), plotted separately for $V_{id} > 0$ and $V_{id} < 0$, to test the exponential weak-inversion model of Section 4.
- Overlay the theoretical tanh curve from Section 10 on the linear plot for direct comparison.

10. Theoretical Transfer Characteristic and Expected Small-Signal Gain

Work these out before wiring up, so results can be checked against theory as they are taken rather than only at the end.

With $I_{SS} = 80 \text{ mA}$ and $\beta \approx 51 \text{ A/V}^2$ (from the BDY's own gfs rating, Section 4), the square-law overdrive works out to $V_{OV0} \approx 40 \text{ mV}$ — confirming weak-inversion operation as discussed in Section 4. Use the exponential model for planning purposes instead:

$$\Delta I_D = I_{SS} \cdot \tanh(V_{id} / 2nVT), \quad g_m = I_{SS} / (nVT), \quad A_d \approx g_m \cdot R_D$$

Taking a typical slope factor $n \approx 1.5$ as a first (planning) estimate gives $g_m \approx 2.06 \text{ A/V}$ and $A_d \approx 268 \text{ V/V}$ ($\approx 48.6 \text{ dB}$) with $R_D = 130 \Omega$ — the largest R_D that keeps the pair out of triode across the full $\pm 300 \text{ mV}$ sweep (about 0.58 V of margin left at the sweep extreme, versus other device-to-device variation in this range). Despite the modest 80 mA bias, A_d comes out large because this device's very high β gives it an unusually large g_m for its operating current — a useful property in ultra-low-power analog design, where a high g_m/I_D ratio at low current is often the design goal. The predicted transfer characteristic and quiescent drain voltages ($V_{D1} = V_{D2} = 6.80 \text{ V}$ at $V_{id} = 0$) are tabulated below for direct overlay on the measured curve:

V _{id} (mV)	ΔI_D (mA)	V _{D1} (V)	V _{D2} (V)
-300	-79.93	11.995	1.605
-200	-79.09	11.941	1.659
-150	-76.73	11.787	1.813
-100	-68.72	11.267	2.333
-50	-45.45	9.754	3.846
-25	-24.93	8.421	5.179
0	0.00	6.800	6.800
25	24.93	5.179	8.421
50	45.45	3.846	9.754
100	68.72	2.333	11.267
150	76.73	1.813	11.787
200	79.09	1.659	11.941
300	79.93	1.605	11.995

n is process-dependent and is not given on the SI9926BDY datasheet, so the estimate above is a planning value only. Section 11 shows how to extract the true n directly from the measured semi-log data and refine this prediction.

11. Calculations — Matching Measured Results with Theory

- On the semi-log plot of $|\Delta I_D|$ versus V_{id} (Section 9), the weak-inversion model predicts a straight-line region; measure its slope and extract $n(\text{measured})$ from it.
- Recompute $g_m(\text{theory}) = I_{SS} / (n(\text{measured}) \cdot VT)$ and $A_d(\text{theory}) = g_m(\text{theory}) \times R_D$ using this measured n , in place of the planning value $n = 1.5$ used in Section 10.
- From the linear ΔI_D - V_{id} plot, determine $g_m(\text{measured})$ directly as the slope at $V_{id} = 0$ (use the two or three points nearest $V_{id} = 0$).

- Compute $A_d(\text{measured}) = g_m(\text{measured}) \times R_D$.
- Fill in the comparison table below and compute the percentage deviation between $g_m(\text{theory, measured } n)$ and $g_m(\text{measured})$. Close agreement confirms that the weak-inversion model correctly describes this device at this bias point.

Quantity	Planning estimate ($n = 1.5$)	Theory with $n(\text{measured})$	From measured slope
g_m (mA/V)	2060	...	...
$A_d = g_m \times R_D$ (V/V)	268	...	...

- Using R_{SS} as the tail resistance, compute $A_{cm} \approx -R_D/(2 \cdot R_{SS})$ and hence $CMRR \approx |A_d(\text{measured})/A_{cm}|$; compare qualitatively with the flatness of the Table 1 data within the ICMR.

12. Precautions

- Do not exceed the maximum V_{DS} (20 V) or V_{GS} (± 12 V) ratings of the SI9926BDY at any point in the experiment.
- Keep the current limit active throughout; do not raise it beyond 150 mA without instructor approval.
- Handle the SO-8 breakout board by its edges; avoid touching the MOSFET pins directly.
- Switch off the supply before altering any connection on the breadboard.